CHAPTER 3

PATHS AND CYCLES

Connectivity

Let us consider four connected graphs of Figure 3.1.
G_1 is a tree, a minimal connected graph; deleting any edge disconnects it. G_2 cannot be disconnected by the deletion of a single edge, but can be disconnected by the deletion of one vertex, its cut vertex.

There are no cut edges or cut vertices in G_3, but even so G_3 is clearly not as well connected as G_4, the complete graph on five vertices.

A vertex cut of G is a subset V' of V such that $G - V'$ is disconnected. A $k-$vertex cut is a vertex cut of k elements.

A complete graph has no vertex cut; in fact, the only graphs which do not have vertex cuts are those that contain complete graphs as spanning subgraphs.

If G has at least one pair of distinct nonadjacent vertices, the connectivity $\kappa(G)$ of G is the minimum k for which G has a $k-$vertex cut; otherwise, we define $\kappa(G)$ to be $v-1$.

Thus $\kappa(G) = 0$ if G is either trivial or disconnected.

G is said to be $k-$connected if $\kappa(G) \geq k$.

All non trivial connected graphs are $1-$connected.

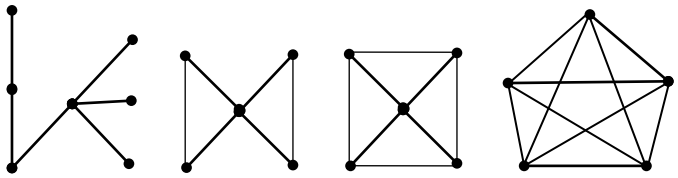
A $k-$edge cut is an edge cut of k elements.

If G is nontrivial and E' is an edge cut of G, then $G - E'$ is disconnected; we then define the edge connectivity $\kappa'(G)$ of G to be the minimum k for which G has a $k-$edge cut.

If G is trivial, $\kappa'(G)$ is defined to be zero.

Thus $\kappa'(G) = 0$ if G is either trivial or disconnected.

$\kappa'(G) = 1$ if G is a connected graph with a cut edge.
G is said to be $k-$edge connected if $\kappa'(G) \geq k$.
All nontrivial connected graphs are $1-$edge connected.



G_1 G_2 G_3 G_4

Figure 3.1

Paths and Cycles

Many Problems can be modeled with paths formed by traveling along the edges of graphs. We begin this section by defining the basic terminology of graph theory that deals with paths.

Definition

- A **walk** is a **finite** sequence of vertices and edges of the form $x_0 \to x_1, \to \ldots, \to x_n$ in which adjacent edges are identical or distinct. Number of edges of a walk is called its length.

- A walk in which all the edges are distinct is called a **trail**.(vertices could be repeated).

- A trail in which all the vertices are distinct is called a **path.**

- A closed path is a **cycle**.

- A closed trail is a **circuit**.

Theorem
There is a simple path between every pair of distinct vertices of a connected undirected graph.

Eulerian and Hamiltonian Graphs

Introduction

The town of Konigsberg, Russia, was divided into four sections by the branches of the Pregel River. These four sections included the two regions on the blanks of the Pregel, Kneiph of Island, and the region between the two branches of the Pregel. In the 18th century seven bridges connected these regions. Figure 4.2(a) depicts these regions and bridges.

The townspeople took long walks through town on Sundays. They wondered whether it was possible to start at some location in the town, travel across all the bridges without crossing any bridge twice, and return to the starting point.

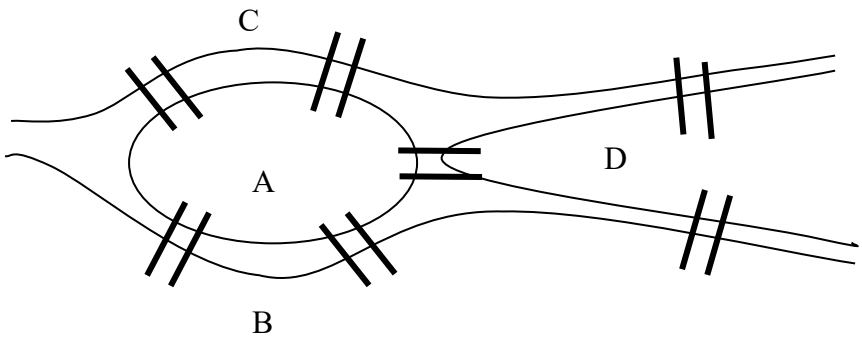


Figure 3.2(a) : The 18th –Century Town of Konigsberg.

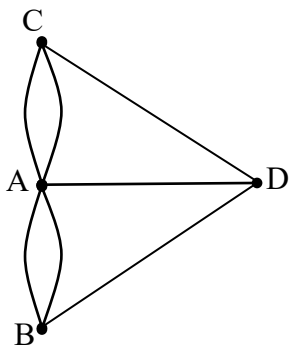


Figure 3.2 (b)

The Swiss mathematician Leonhard Euler solved this problem. His solution, published in 1736, may be the first use of the graph theory. Euler studied this problem using the multigraph. Obtained when **4 regions are represented by vertices and the bridges by edges**. This multigraph is shown in Figure 4.3 (b).

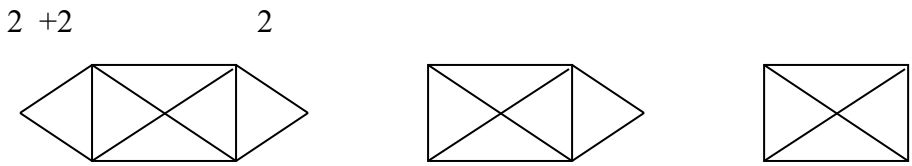
The problem of traveling across every bridge without crossing any bridge more than once can be rephrased in terms of this model. The question becomes: Is there a simple circuit in this multigraph that contains every edge?

Eulerian Graph

A connected graph G is <u>Eulerian</u> if there exists a closed trail containing every edge of G. Such a trail is an <u>Eulerian trail</u>.

Note that this definition requires each edge to be traversed once and once only.

A non-Eulerian graph G is <u>semi-Eulerian</u> if there exists a trail containing every edge of G. The following are Euerian, semi-Eulerian and non-Eulerian graphs respectively.



Eulerian Semi-Eulerian Non-Eulerian

Figure 3.4

Theorem 1

A graph G is bipartite if and only if each cycle of G has even length.

Proof: $\Rightarrow$ Let $G = (V(G), E(G))$ be a bipartite graph.

i.e. $V(G) = V_1 \cup V_2$ and each edge joins an vertex of V_1 with a vertex of V_2.

Now take any cycle of G, say, $v_1 \to v_2 \to \cdots v_m \to v_1$ (edges and vertices are distinct) of length m.

With out loss of generality, we may assume that $v_1 \in V_1$.

Then
$$\begin{array}{ccccccccccc} v_1 & \to & v_2 & \to & v_3 & \to & v_4 & \to & \cdots\cdots v_m & \to & v_1 \\ \in & & \in & & \in & & \in & & \in & & \in \\ V_1 & & V_2 & & V_1 & & V_2 & & V_2 & & V_1 \end{array}$$

$\therefore m$ must be even.

$\Leftarrow$ Let G be a graph whose cycles are all of even length. Without loss of generality we may assume that G is connected.

Take any vertex, say $v_1 \in V$, and let

$$V_1 := \{u \in V \mid d(v_1, u) \quad \text{is even}\} \cup \{v_1\}$$
$$V_2 := V \setminus V_1$$

so we have, $V = V_1 \dot{\cup} V_2$ and any edge joins a vertex of V_1 with a vertex V_2.

Lemma

If G is a graph in which the degree of each vertex is at least 2, then G contain a cycle.

Proof

If G has any loop or multiple edges, then the result is trivial.
Suppose G is a simple graph. Let v be any vertex of G. Choose v_1 to be any vertex adjacent to v. So that for each $i > 1$ choose v_{i+1} to be any vertex adjacent to v_i except v_{i-1}.

Then construct a walk $v \to v_1 \to v_2 \to \cdots v_{i-1} \to v_i \to v_{i+1}$.

Since G has only finitely many vertices we must eventually choose a vertex that has been chosen before.

If v_k is the first such vertex, then that part of the walk lying between the two occurrences of v_k is the required cycle.

Theorem 2

A connected graph *G* is Eulerian if and only if the degree of each vertex of *G* is even.

Proof:

$\Rightarrow$ Suppose that P is an Eulerian trail of G.
Whenever P passes through a vertex then there is a contribution of 2 towards the degree of that vertex. Since each edge occurs exactly once in P, each vertex must have even degree.

$\Leftarrow$ Prove by induction on the number of edges of G.(i.e. Assume that the assertion is true for the graphs whose number of edges are less than number of edges of G).
Suppose the degree of each vertex of G is even.

Since G is connected and each vertex has degree at least 2, by the Lemma 1, G contains a cycle C. If C contains every edge of G, then the proof is completed.

If not, we remove C from G, and form a new possibly disconnected graph H, where $|E(H)| < |E(G)|$, in which each vertex of H still has even degree.

By the induction hypothesis, each component of H has an Eulerian trail.

Since each component of H has at least one vertex in common with C, by connectedness, we have required Eulerian trail of G by following the edges of C until a non-isolated vertex of H is reached, tracing the Eulerian trail of the component of H which contains that vertex, and then continuing along the edges of C until we reach a vertex belonging to another component of H and so on. (see Figure 4.5)

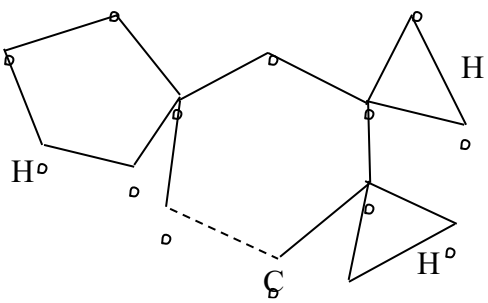


Figure 3.5

Corollary 1

A connected graph is Eularian if and only if its set of edges can be split up into disjoint cycles.

Corollary 2

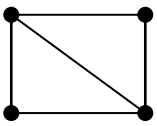
A connected graph is semi-Eularian if and only if it has exactly two vertices of odd degree.

Hamiltonian Graph

We have developed necessary and sufficient conditions for existence of paths and cycles that contain every edge of a multigraph exactly once.
Here we determine whether there exists a closed trail passing exactly once through each vertex of G.
Note that such a trail must be a cycle (except when G is a null graph). Such a cycle is a Hamiltonian cycle and G is a Hamiltonian graph.

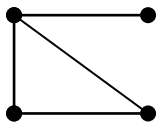
A non-Hamiltonian graph G is <u>semi-Hamiltonian</u> if there exists a path passing through every vertex.

The following are Hamiltonian, semi-Hamiltonian and non-Hamiltonian graphs respectively.



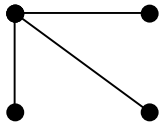
Hamiltonian

Ha cycle



Semi-Hamiltonian

Ha path



Non-Hamiltonian

neither

Theorem 3 (Ore, 1960).
If G is a simple graph with $n(\geq 3)$ vertices, and if $\deg(v)+\deg(w)\geq n$ for each pair of non-adjacent vertices v and w, then G is Hamiltonian.

Examples

1. A connected graph need not have a Hamiltonian path. eg. Tree.
2. K_2 has non-cyclic Hamiltonian path; if $n>2$, K_n has a Hamiltonian cycle.
3. For a graph G with n vertices, $n\geq 3$; having a Hamiltonian circuit is equivalent to having a cycle C_n as a sub graph.

Remark
If all the vertices have sufficiently high degrees then a Hamiltonian path exists.

e.g. If G has n vertices, and degree of each vertex is at least $\frac{n}{2}$, then there is a Hamiltonian cycle.

Corollary 1

If G is a connected simple graph with n vertices where $n\geq 3$, then G has a Hamiltonian cycle if the degree of each vertex is at least $\frac{n}{2}$.

Applications

Many problems can be modeled using graphs with weights assigned to their edges. Problems involving distances can be modeled by assigning distances between cities to the edges, problems involving flight time can be modeled by assigning travel time between cities to the edges and problems involving traveling cost can be modeled by assigning traveling cost between cities to the edges. Graphs with weights assigned to their edges are called *weighted graphs*.
Weighted graphs are used to model computer networks. Communication costs, the response times of the computers over these lines, or the distance between computers, can also be studied using weighted graphs.
Several types of problems involving weighted graphs arise frequently. Determining the path of least length between two vertices in a network is one such problem.

The *length* of a path in a weighted graph is the sum of the weights of the edges of the path. Then the path of least length, between two given vertices is the shortest path.

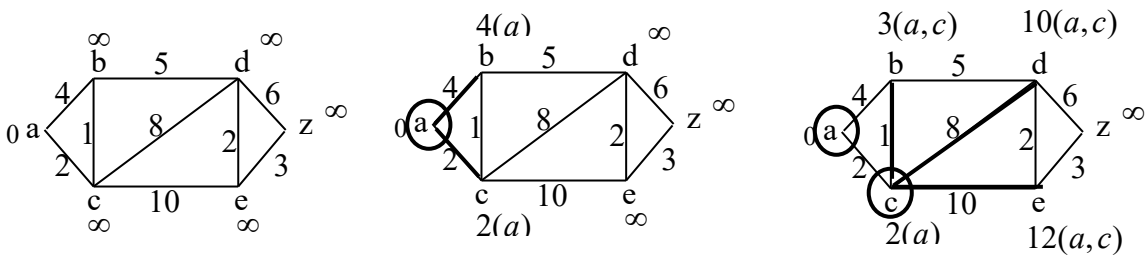
Shortest Path Problem

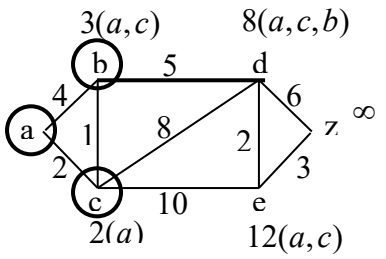
There are several different algorithms that can be used to find the shortest path between two vertices in a weighted graph.

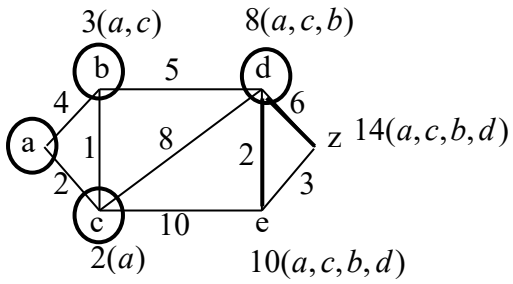
An algorithm discovered by the Dutch Mathematician E. Dijkstra, in 1959, is called Dijkstra's Algorithm.

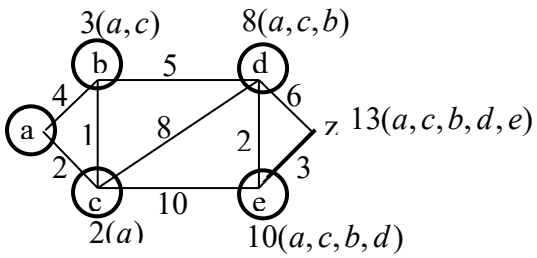
Example 1

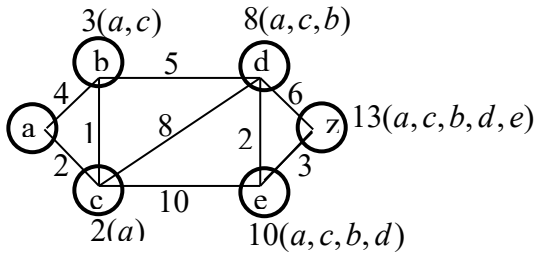
Find the shortest path from **a** to **z** of the following weighted graph.
Label vertex a with 0 and all the other vertices with ∞. Use notation $L_0(a)=0$, $L_0(v)=\infty$, for these vertices before any iterations have been taken place. In each iteration circle the vertex of smallest labeling.









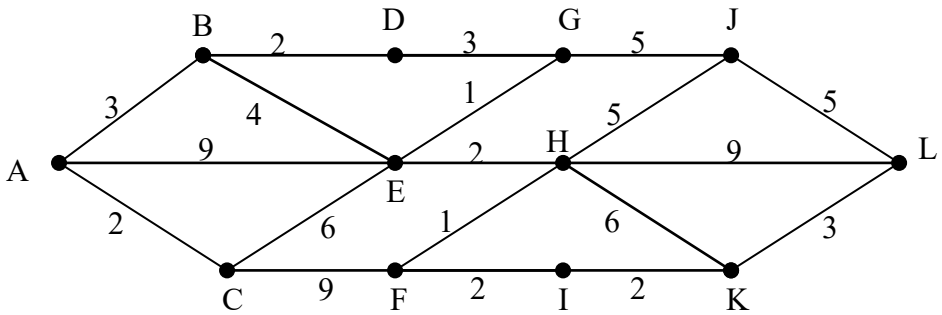


$\therefore$ The Shortest path from **a** to **z** is a,c,b,d,e of length 13.

Alternating Method

Suppose that we have a 'map' of the form shown below, in which the letters *A*-*L* refer towns that are connected by roads. If the lengths of these roads are as marked, what is the length of the shortest path from *A* to *L*?

Note that the numbers in the diagram need not refer to the lengths of the roads, but could refer to the times taken to travel along them, or the costs of doing so. Thus, if we have an algorithm for solving this problem in its original formulation, then this algorithm can also be used to give the quickest or cheapest route.



Note that the upper bound for the answer can easily be obtained by taking any path from A to L and calculating its length. For example, the path $A \to B \to D \to G \to J \to L$ has total length 18, and so the length of the shortest path cannot exceed 18.

Method of Solving

- Move across the graph from left to right.
- First assign temporary labeling $L_t(V)$ to each vertex V, which is the distance from A to V.
- Then make permanent the smallest of the temporary labeling.

Take $L(A) = 0$.

<u>Step 1</u>

Label all the vertices adjacent to A.

$L_t(B) = L(A) + 3 = 0 + 3 = 3$.

$L_t(E) = L(A) + 9 = 0 + 9 = 9$.

$L_t(C) = L(A) + 2 = 0 + 2 = 2$.

Make permanent C; $L(C) = 2$.

<u>Step 2</u>

Label all the vertices adjacent to C.

$L_t(E) = L(C) + 6 = 2 + 6 = 8.$

$L_t(F) = L(C) + 9 \ = 2 + 9 \ = 11.$

List all the temporary labeling;

$L_t(B) = 3$, $L_t(E) = 8$, $L_t(F) = 11$.

Make permanent B; $L(B) = 3$.

<u>Step 3</u>

Label all the vertices adjacent to B.

So the Shortest path has length 17 and the path is $A \to B \to E \to H \to F \to I \to K \to l.$

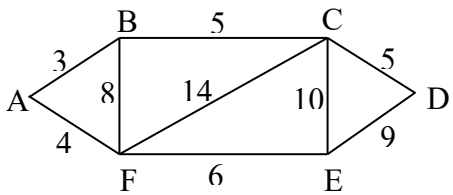
The Chinese Postman Problem

A postman wishes to deliver letters, covering the least possible total distance and returning to his starting point. He must obviously traverse each road in his route at least once, but should avoid covering too many roads more than once.

This problem can be reformed in terms of a weighted graph, where the graph corresponds to the network of roads, and the weight of each edge is the length of the corresponding road. In this reformulation, the requirement is to find a closed walk of minimum total weight that includes each edge at least once. If the graph is Eulerian, then any Eulerian trail is a closed walk of the required type. Such an Eulerian trail can be found by Fleury's algorithm. If the graph is not Eulerian, then the problem is much harder, although an efficient algorithm is known. To illustrate the ideas involved we look at special types of graphs, in which exactly two vertices have odd degrees.

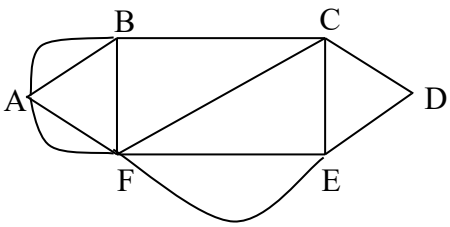
Example 2

A postman collects his mail from one of the locations A,B,C,D,E deliver his mail and return to the starting location. Find the minimum distance he has to traverse in order to deliver all the mail. Also, at which locations he has to start his duties in order to complete the entire job in traveling minimum distance.



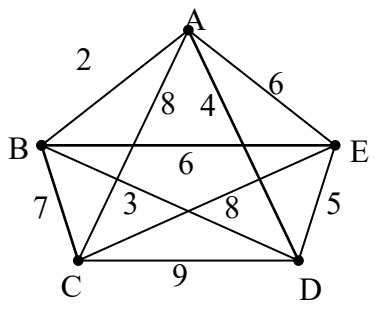
Since vertices B and E are the only vertices of odd degrees, we can find a semi-Eulerian trail from B to E covering each edge exactly once. In order to return to the staring point, covering the least possible distance, we now find the shortest path from E to B using one of the algorithms described above.

The solution of the Chinese postman problem is then obtained by taking this shortest path $E \to F \to A \to B$, together with the original semi-Eulerian trail, giving a total distance of $13 + 64 = 77$. Note that, if we combine the shortest path and the semi-Eulerian trail, we get an Eulerian graph.



Traveling Salesman Problem

Example 3

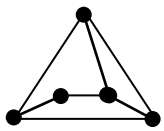


A traveling salesman wishes to visit several given cities and return to his starting point, covering the least possible total distance. For example, suppose there are five cities A, B, C, D and E, and the distances between the cities are given in the following figure.

To solve this problem we have to find closed cycle of least weight containing all the vertices. Which equivalent of finding Hamiltonian cycle with least weight. For example, suppose there are five cities A, B, C, D and E and the distances between the cities are given as in the following figure. To find the solution to Traveling Salesman problem, we have to find a Hamiltonian cycle of least weight, which is $A \to B \to D \to E \to C \to A$ of total 26.

Assignment 3

3.1 (i) A graph G is shown below. Draw its complement $\overline{G}$.



(ii) What are the lengths of all possible closed paths in G staring at vertex a

(iii) Either find an Euler cycle in the graph G of part (i), or explain why no such cycle exists.

(iv) Either find a noncyclic Euler path in G, or explain why no such path exists.

(v) Either find a Hamiltonian circuit in G, or explain why no such circuit exists.

(vi) Either find a noncyclic Hamiltonian path in G, or explain why no such path exists.

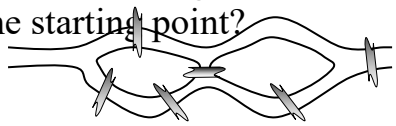
3.2 In the Peterson graph, find

(i) a trail of length 5

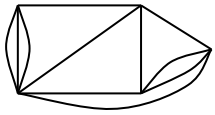
(ii) a path of length 9

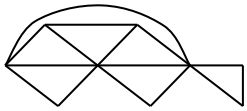
(iii) a cycle of length 5,6,8 and 9.

3.3 Can someone cross all the bridges shown in the following map exactly once and return to the starting point?

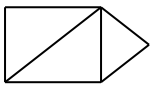


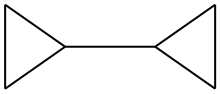
3.4 Determine whether the given graph has an Euler trail. If it does, find such a trail.





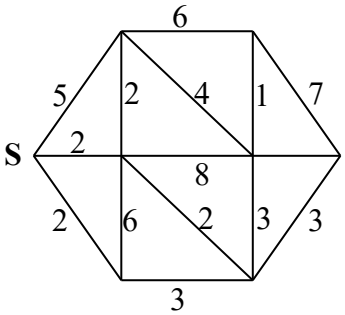
3.5 Determine whether the given graph has a Hamiltonian circuit. If it does, find such a circuit.





3.6 Find the shortest path from S to each other vertex in the weighted graph.

(a)



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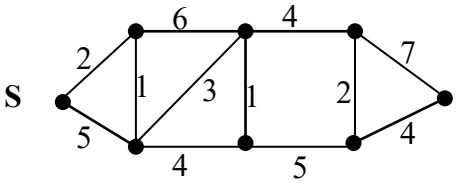
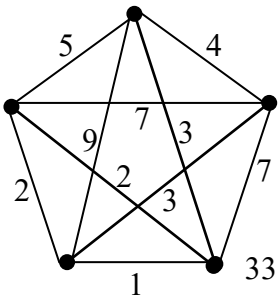
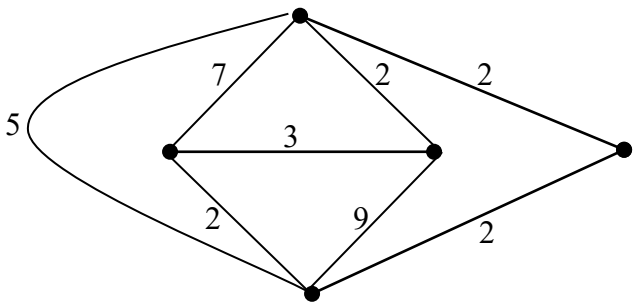


3.7 Solve the Traveling Salesman problem for the weighted graph.



3.8 Solve the Chinese Postman problem for the weighted graph.



3.9 When can the center lines of the streets in a city can be painted without traveling a
street more than once. (Assume that all the streets are two-way streets)

3.10 For which values of n do the following graphs have an Euler trail?

(a) K_n (b) C_n (c) W_n (d) Q_n

3.11 For which values of n and m does the complete bipartite graph $k_{m,n}$ have a Hamiltonian circuit?

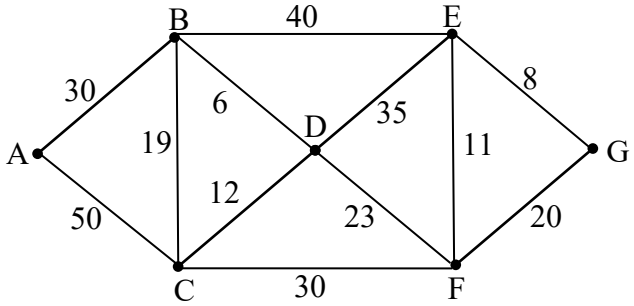
3.12 Show that a bipartite graph with an odd number of vertices does not have a Hamiltonian circuit.

3.13 Define an Euler trail and a Hamiltonian cycle. Determine which of the following graphs are Eulerian and which are Hamiltonian:

(i) The graph of cube
(ii) The Peterson graph
(iii) The complete graph K_5
(iv) The wheel graph W_6

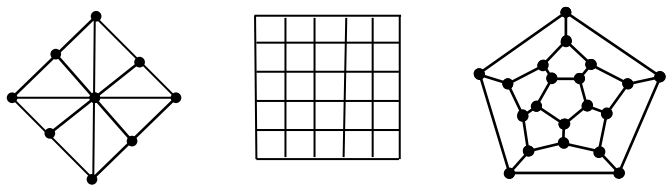
3.14 If G is bipartite, then prove that each cycle of G has even length.

3.15 Find the shortest path from A to G of the following graph.



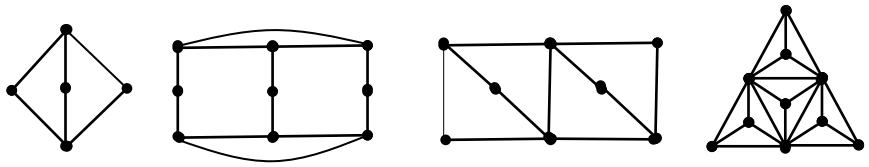
3.16 (a) Draw a graph with a Hamiltonian circuit but no Euler circuit.
(b) Draw a graph with an Euler circuit but no Hamiltonian circuit.

3.17 Find a Hamiltonian circuit in each of the following graphs.

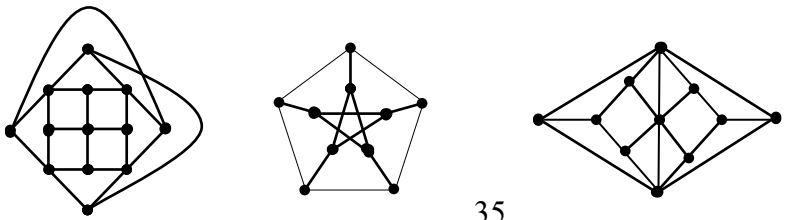


(a) (b) (c)

3.18 Find a Hamiltonian path in each of the following graphs and prove that no Hamiltonian circuit exists:

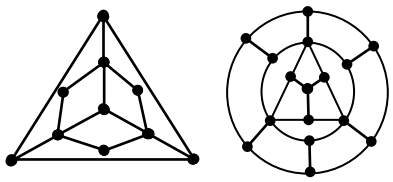


(a) (b) (c) (d)



(e) (f) (g)

3.19 Prove that the following graphs have no Hamiltonian circuits:

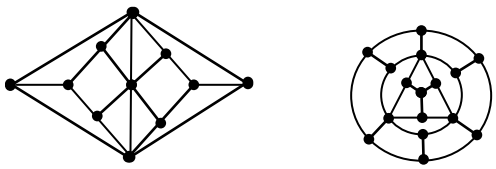


(a) (b)

3.20 If a connected graph has no circuits of odd length, then it is possible to uniquely "color" the vertices red and blue such that every edge goes between vertices of different colors. Such graphs are called bipartite.

(i) Show that if a connected bipartite graph has a Hamiltonian circuit, then the numbers of red and blue verticals must be equal; if it has a Hamiltonian path, then the numbers of reds and blues can differ by at most one.

(ii) Use part (i) to show that the following graphs have no Hamiltonian circuit:



(a) (b)